

Questions

1.) SQL Joins

1) Retrieve all employees and their assigned projects using an Inner Join

Select A.EmployeeID, A.FirstName, A.LastName, A.Department, A.Salary, B.ProjectID,
B.ProjectName, B.Budget, B.Status

From employees as A

Inner join projects as B

On A.EmployeeID = B.EmployeeID

First Name	Last Name	Department	Salary	Project ID	Project Name	Budget	Status
John	Doe	IT	10000	101	AI Development	100000	Completed
John	Doe	IT	10000	101	Cybersecurity Audit	15000	Pending
Alice	Smith	HR	60000	102	Employee Training	50000	Ongoing
Bob	Johnson	Finance	75000	104	Financial Analysis	90000	Ongoing
Emma	Wilson	Sales	65000	105	Market Expansion	65000	Completed
Michael	Clark	Finance	80000	106	Risk Management	80000	Pending

2) Retrieve all employees and their assigned projects, including employees who have no projects using a left join.

Select A.EmployeeID, A.FirstName, A.LastName, A.Department, A.Salary, B.ProjectID,
B.ProjectName, B.Budget, B.Status

From employees as A

left join projects as B

On A.EmployeeID = B.EmployeeID

Exercise 5

Table 1: Employees

Q1) Show all employees with their salary. If salary is Null, display 0

|| Select employee_id, name, ~~salary~~ IFNull(salary, 0) as salary-with-default
from employees

employee_id	name	salary-with-default
1	Alice	5000
2	Bob	0
3	Charlie	1000
4	Diana	0

Q2) Show employee names with their department. If department is Null, show "Not Assigned".

|| Select employee_id, name, ~~department_name~~ IFNull(department, 'Not Assigned') as department
from employees

employee_id	name	department
1	Alice	HR
2	Bob	IT
3	Charlie	Not Assigned
4	Diana	Finance

Table 2: Orders

Q3) Find orders with Null customer_id using IS Null

// Select order_id, customer_id
from orders
where customer_id is Null

order_id	customer_id
103	Null

Q4) Show all orders. If delivery_date is Null, show 'Pending'

// Select order_id, customer_id, ~~delivery_status~~ If Null (delivery_status, 'Pending')
from orders.

order_id	customer_id	delivery_status
101	101	2024-11-20
102	202	Pending
103	Null	2024-11-20

Table 3: Students

Q5 Show all students and their grades. Replace Null with 0

// Select student_id, name, IfNull(grade, 0)

From Students

student_id	name	grade
1	Gihan	85
2	Mary	0
3	Omar	90

Q6 Count students who haven't been graded.

// Select count(student_id) as count-Student, grade.

from students

where grade is Null

count-Student	grade
1	Null

Table 4: Products

Q1 Show product name, price and final price after discount (assume 0 if discount is Null)

// Select product_id, name, (price - If Null(discount, 0)) as final_price
from products

product_id	name	final_price
501	Keyboard	25
502	Mouse	10
503	Monitor	100

Table 5: customers

Q2 Count how many customers have no email

// Select count(name) as missing_email_count
from customers
where email is Null

missing_email_count
2

Q3 Show all customers with customers with email. If Null, display "No email".

// Select If Null(email, "No email") as email_display, name, customer_id.
from customers

email_display	name	customer_id
No email	Linda	1
joseph@email.com	Joseph	2
No email	Wick	3

Table 6: Payments

Q10) Show payment details with method replaced by "Unknown" if Null

// Select payment_id, IfNull(method, "Unknown") as method_display, status

payment_id	method_display	status
301	Credit	Null
302	PayPal	Success
303	Unknown	Failed

Table 7: Inventory

Q11) Show items and their quantity (0 if Null)

// select item_id, item_name, IfNull(quantity, 0) as quantity_checked
from Inventory

item_id	item_name	quantity_checked
1	Pen	0
2	Notebook	150
3	Erasor	0

Table 8: Employees - Extra

Q12) Show employee ID and the first available value among bonus or commission

Select coalesce(bonus, commission, 0) as ^{available_reward}~~amount~~, emp_id

available_reward	emp_id
300	1
100	2
0	3

Table 9: classes

Q13) Count classes that don't have a room assigned

11 Select count(room room) as no_room_count
from classes
where room is Null

no_room_count
2

Q14) Show attendance records with status. Replace Null with "Not Marked"

11 Select student_id, date, IFNull(status, "Not Marked") as attendance_status
from Attendance

Student_id	date	attendance_status
1	2025-04-01	Not Marked
2	2025-04-01	Present
3	2025-04-01	Absent

Table 11: Bank_accounts

Q15) Show account ID, account_type (or 'Unknown') and balance (or 0)

11 Select account_id, IFNull(account_type, 'Unknown') as type_display, IFNull(balance, 0) as
balance_checked
from Bank_Accounts

account_id	type_display	balance_checked
A1	Savings	0
A2	Current	5000
A3	Unknown	2000

Table 12: projects

Q16) Show all projects with a start date. If start_date is Null, display 'TBD'.

Select project_id, title, IFNULL(start_date, 'TBD') as start_display.
From projects

project_id	title	start_display	start_display
1	Website Revamp		2025-01-10
2	Mobile App		TBD
3	Data Migration		TBD

Table 13: reviews

Q17) Display reviews showing comment (or 'No comment') and rating (or 0)

Select review_id, product_id, IFNULL(comment, 'No comment') as comment_display,
IFNULL(rating, 0) as rating_display
From reviews

review_id	product_id	comment_display	rating_display
1	501	Great product	4
2	502	No comment	0
3	503	Works fine	3

Table 14: Suppliers

Q14) Show the supplier contact number. Use Coalesce (phone, alt_phone, 'No contact')

Select supplier_id, name, coalesce(phone, alt_phone, 'No contact') as contact_number
From Suppliers

Supplier_id	name	contact_number
1	Global Goods	123456789
2	Best Supplies	987654321
3	Value Source	No contact.

Table 15: User_settings

Q15) Show all users and their preferences. Replace all Nulls with defaults.

as theme_set
Select user_id, IFNull(theme, "light") as theme_set, IFNull(language, "English") as language_set,
IFNull(timezone, "UTC") as timezone_set
From user_settings

user_id	theme_set	language_set	timezone_set
1	light	English	UTC
2	Dark	English	UTC + 1
3	light	English	UTC

Table 16 : Maintenance

Q10) Show maintenance log with

select record_id, machine_id, IfNull(issue, "Unknown issue") as issue_log,
IfNull(technician, "Not assigned") as technician_name
from maintenance

record_id	machine_id	issue_log	technician_name
1	M101	Overheating	Not Assigned
2	M102	Unknown issue	Not Assigned
3	M103	Jammed	Alex